MILITARY SPECIFICATION

CLAMP, COARCTATION, GLOVER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a coarctation clamp, of the Glover design, suitable for use in cardiovascular surgical procedures.

1.2 Classification. Clamps shall be of the following type as indicated below (see 6.2):

Type		
1	-	Straight
2	-	Angular

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-L-10547	- Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

2.1.2 Drawing. The following drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DRAWING

Defense Personnel Support Center (DPSC)

22251 - Clamp, Coarctation, Glover, 8-3/4 Inches (22.5 cm)

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

F 899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

F 921 - Standard Deviation of Terms Relating to Hemostatic Forceps

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The clamps shall be in accordance with the applicable requirements of GG-I-526 and the terms used to describe the components and portions of the instruments described in this specification shall be in accordance with ASTM F921 and as specified herein.

3.2 Material. The clamps shall be fabricated from a corrosion-resistant stainless steel class 4, in accordance with F899.

3.3 Hardness. After proper stress relieving, the Rockwell hardness shall be RC 40 to RC 45.

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3.4 Style, design and dimensions. The clamps shall be of the Glover design, suitable for use in cardiovascular surgical procedures. The clamps shall have a jaw section of 5.5 cm nominal length. One jaw shall be provided with two rows of finely serrated teeth separated by a uniform space. The other jaw shall have a single, central row of finely serrated teeth. The width of the rows of teeth, the spacing between the rows on the two-row jaw, and the depth of interdigitation of the rows shall be matched to provide a firm, non-slip, grasp, without trauma to the tissue. The clamps shall be of the finger ring type with ratchet. The shape and style of the clamps shall be in accordance with DPSC Drawing No. 22251.

3.5 Locks. The clamps shall have a box type lock. The box lock construction shall be accurately fitted and shall be without cracks, burrs or sharp edges at any place in the lock construction. The pin or rivet shall be permanently secured. The joint performance shall be smooth, of equal resistance and nonbinding in all positions when opening or closing the clamps. The lock may be lubricated. Under no circumstances shall looseness or free play of either shank permit any misalignment of the jaws in closing. NOTE: This paragraph shall replace the requirements specified in GG-I-526 for locks.

3.6 Finish. Finish shall be mirror or satin. Should the instrument be of the mirror finish, the locks and ratchets shall be polished smooth, but need not possess a mirror finish.

3.6.1 Passivation. The clamps shall be properly passivated.

3.7 Jaw and ratchet operation. When tested as specified in 4.4.2, the jaw tips shall close first and progressively tighten with each ratchet position. Each ratchet position shall hold firmly and release instantly.

3.8 Identification markings. In addition to the requirements specified in GG-I-526, clamps of foreign manufacture shall be marked "stainless."

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection. Inspection shall be in accordance with GG-I-526.

4.1.3 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.4 Certificate of quality. Certificate of quality supplied by the manufacturer of the clamp may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purposes shall be one clamp.

TABLE I. Sampling criteria for examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purposes shall be one clamp. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

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TABLE II. Sampling criteria for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Hardness test	3.3 & GG-I-526	4.4.1	S-2	1.5
Dimensions	3.4	---	S-2	2.5
Copper sulfate test	GG-I-526	GG-I-526	S-2	1.0
Boil test	GG-I-526	GG-I-526	S-2	1.0
Jaw and ratchet operation test	3.7	4.4.2	S-2	1.0

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the manufacture of the clamp shall be selected for chemical analysis.

4.3 Examination.

4.3.1 Examination of clamps. The clamps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any clamps in the sample containing one or more defects shall be rejected and if the number of defective clamps in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	Clamps not of the specified design for appropriate type, stock number and item identification.
102	Both jaws not serrated throughout full length.
103	Jaw serrations not of uniform depth and not well defined.
104	Jaw serrations not chamfered throughout full length.
105	Ratchet teeth not well defined.
106	Clamps not free from cracks, pits, burrs, sharp edges, scale, rough areas or corrosion.
107	Finish of clamps not uniform and not as specified.

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TABLE III. Classification of defects (Cont'd)

Categories	Defects
Major	
108	Serrations of both jaws do not completely mesh in fully closed position.
109	Box lock performance not as specified in 3.5.
110	Jaws possess sharp tips.
111	Ratchet teeth do not completely engage in all positions.
112	Pin for box lock not flush with surface and not free from crevices.
Minor	
201	Clamps not free from dirt, polishing material or foreign matter.
202	Identification markings not present, not correct, not complete, or not legible.

4.4 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

4.5.2 Jaw and operation test. The clamps shall be positioned to the first ratchet position, but not engaged. Clamps with jaw tips not touching shall be rejected. Further, should all the jaw serrations touch, the clamps shall be rejected. The clamps shall then be engaged to each of the ratchet positions. Clamps that do not progressively tighten with each ratchet advance shall be rejected.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

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TABLE II. Sampling criteria for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Hardness test	3.3 & GG-I-526	4.4.1	S-2	1.5
Dimensions	3.4	---	S-2	2.5
Copper sulfate test	GG-I-526	GG-I-526	S-2	1.0
Boil test	GG-I-526	GG-I-526	S-2	1.0
Jaw and ratchet operation test	3.7	4.4.2	S-2	1.0

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the manufacture of the clamp shall be selected for chemical analysis.

4.3 Examination.

4.3.1 Examination of clamps. The clamps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any clamps in the sample containing one or more defects shall be rejected and if the number of defective clamps in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	Clamps not of the specified design for appropriate type, stock number and item identification.
102	Both jaws not serrated throughout full length.
103	Jaw serrations not of uniform depth and not well defined.
104	Jaw serrations not chamfered throughout full length.
105	Ratchet teeth not well defined.
106	Clamps not free from cracks, pits, burrs, sharp edges, scale, rough areas or corrosion.
107	Finish of clamps not uniform and not as specified.

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5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Clamps packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Clamps packed as specified in 5.2.3 shall be unitized as specified in 5.2.5.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.4.

6. NOTES

6.1 Intended use. The coarctation clamp shall be used for cardiovascular surgical procedures.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) National Stock Number.

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5.1.1 Level A.

5.1.1.1 Cleaning and drying. Clamps shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each clamp shall be packaged in accordance with MIL-P-116, Method IC-1.

5.1.1.4 Intermediate package. Twelve unit packages containing clamps of one type only shall be packaged in a box conforming to PPP-B-566, PPP-B-636, Class domestic or PP-B-676.

5.1.2 Level C. The level C preservation for clamps shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twelve (12) intermediate packages containing clamps of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601 overseas type, PPP-B-621, class 12 or PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twelve intermediate packages containing clamps of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 2 or PPP-B-636 class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. The level C packing for clamps shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-37976	CLAMP, COARCTATION, GLOVER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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